DATA SATURDAYS

DATA SATURDAY SOFIA 2026

When Enterprise Data Disagrees

Reconciliation Patterns in Microsoft Fabric

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Meet the speaker

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APIs, data integration and systems people use every day

An address exists. Its street link is missing

What would you check before using this address?

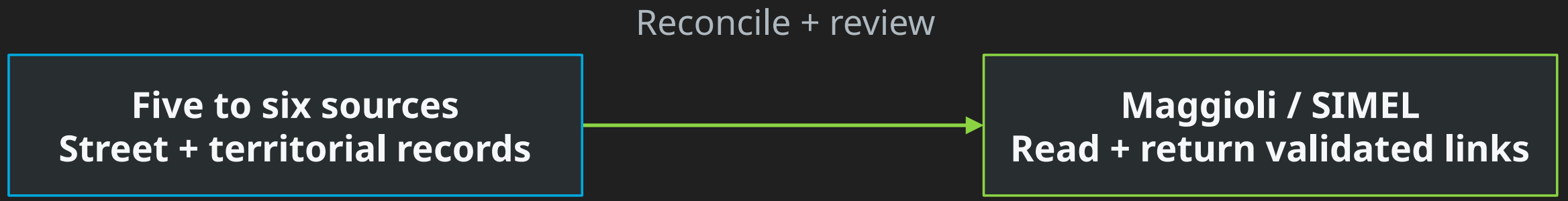
Maggioli / SIMEL	Office address feed
MAG:ODN:000184	CIV:ADR:012784
A. Celi Street	Adolfo Celi Street, 253
Street reference available	**Street link: MISSING**

Keep the record and check its street reference.

THE MESSINA STORY

Let's start with a real project

Municipality of Messina



My role: architecture, full-stack development and production delivery

Messina: territorial data in daily work

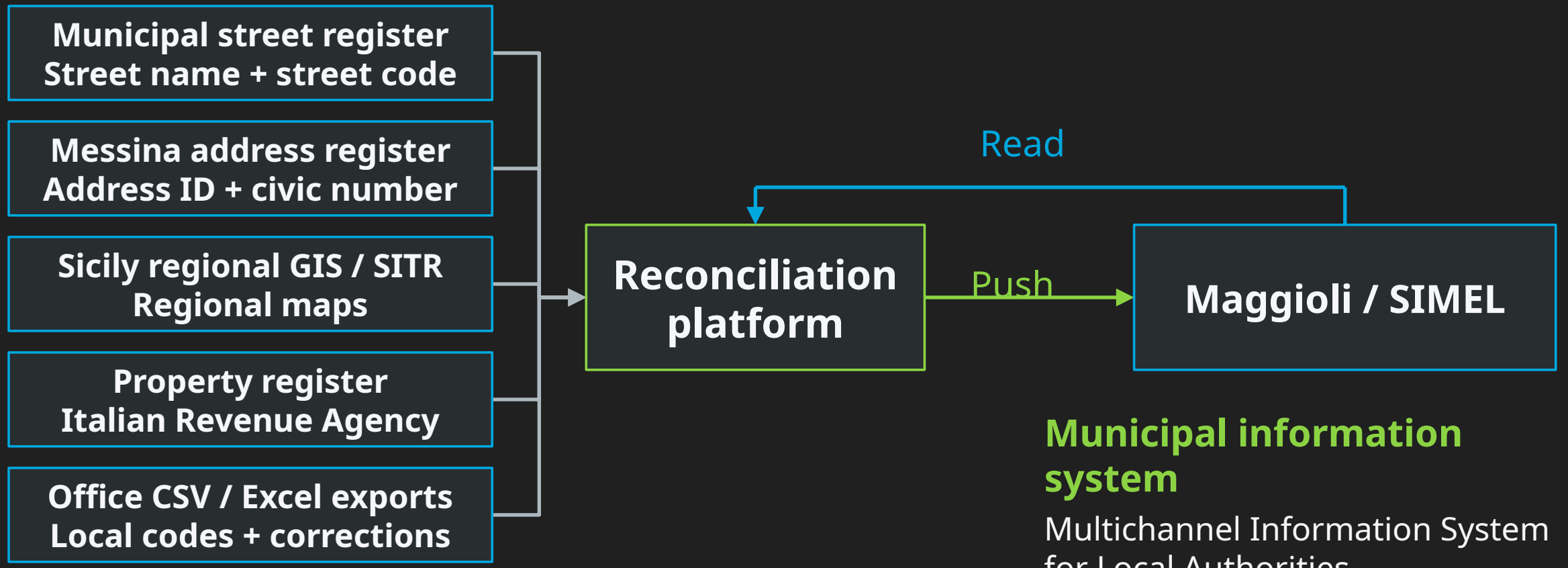


Street-name reconciliation and territorial data

Architecture, full-stack development and production delivery

Photo: Jeroen P, CC BY 2.0. Cropped.

Messina: the integration pattern



Municipal information system

Multichannel Information System for Local Authorities

A system, a dataset and an extract have different roles.

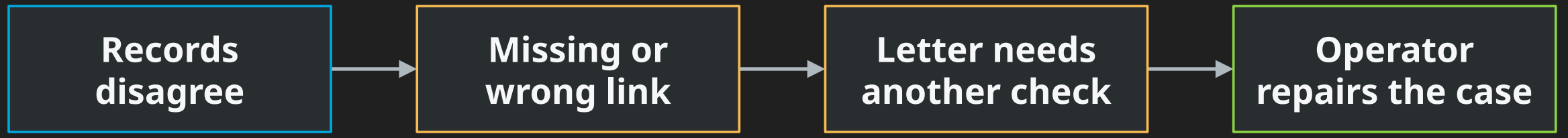
Similar names can hide a different object

Source key	Original name	Area type
MAG:ODN:000184	**A. Celi Street**	Street
TOP:STR:000531	**Adolfo Celi Street**	Street
OFF:ROW:000009	Adolfo Celi Square	**Square**

Abbreviation: candidate. Different area type: investigate.

Keep the area type separate from the street name.

A broken link becomes someone’s extra work

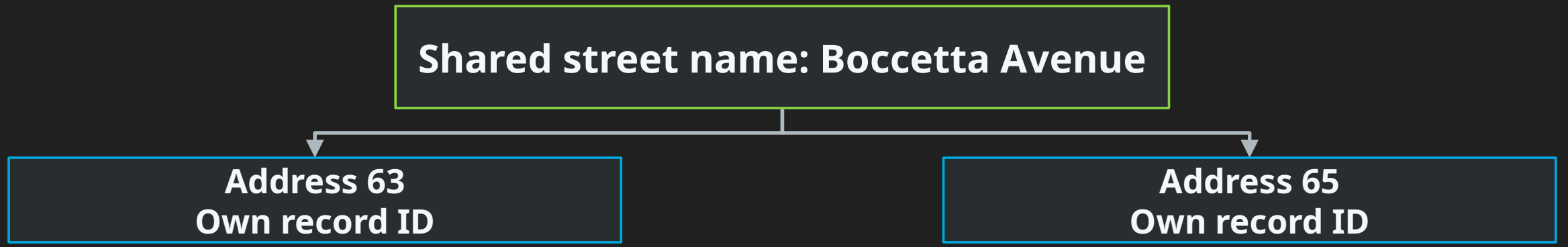


Process	Possible consequence
Letter / fine notice	Wrong destination or another check
Service / territorial case	No usable address or wrong context
Map / address count	Duplicates change the result
Correction stays local	Next process still uses the old reference

Possible impact scenarios, not documented incidents in Messina.

One street, two different addresses

Address ID (short)	Street name	Civic number
019e5e6c…	Boccetta Avenue	63
019e5e66…	Boccetta Avenue	65



Municipality of Messina, CC BY 4.0; © OpenStreetMap contributors. Street context only.

Search finds a candidate; evidence supports the link

Query: A. Celi Street



PostgreSQL + pg_trgm

FastAPI
Candidates + source keys

PostGIS
Geographic context

Near in space and similar in text are clues to investigate

Geographic comparisons need the right units: degrees and metres differ.

The evidence behind a validated street link

Evidence	Source record	Candidate street
Source key	MAG:ODN:000184	TOP:STR:000531
Name	A. Celi Street	Adolfo Celi Street
Area type	Street	Street

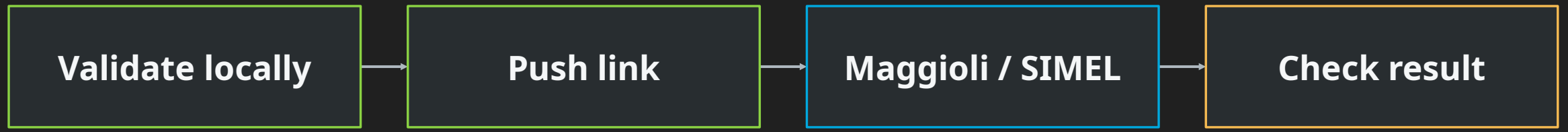
Name + area type: candidate

Checked reference: both records refer to the same street

REF-CELI-01: approve the street link, keep address identities

Validate the street link. Keep the evidence and both source keys.

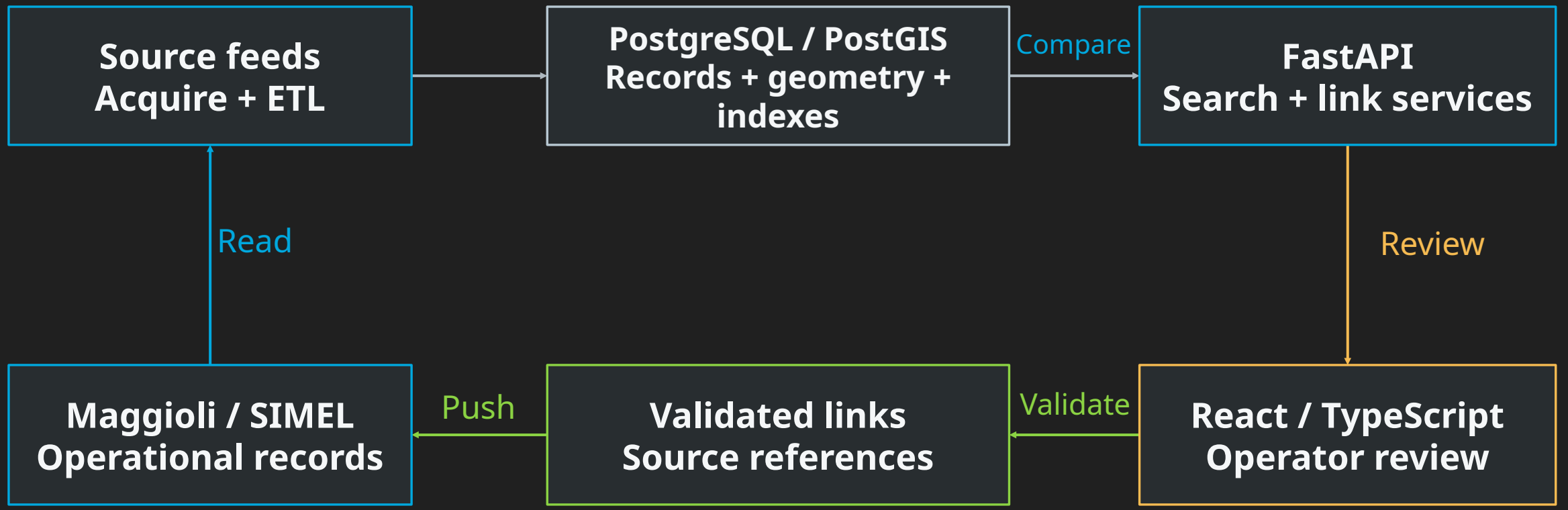
Validated locally, still missing at the destination



Local platform	Receiving system
MAG:ODN:000184	MAG:ODN:000184
TOP:STR:000531	Street link still missing
VALIDATED LOCAL	**TARGET PENDING**

Proposed integration pattern: check acceptance and retry safely under the real contract.

The engineering behind the reconciliation loop



Docker deployment and production stabilisation. Different business, same problem.

WHEN ENTERPRISE DATA DISAGREES

Same customer, different records

The object changes. The need for evidence remains.

CRM:C101 Alfa Ltd	ERP:E501 ALFA LTD	PORTAL:P901 Alfa

One company? We need evidence.

Same customer, different records

Source key	Company	Company ID	Address
CRM:C101	Alfa Ltd	DEMO-001	Roma Street 12
ERP:E501	ALFA LTD	DEMO-001	Roma St., 12
PORTAL:P901	Alfa	DEMO-001	**Roma Street 10**

CRM: sales view **ERP: billing view** **API: record extract**

We are matching companies

Sites, contacts and accounts have their own identities.

The business cost of a wrong customer link

One company stays split

Support sees only part
of the customer history

Different companies are joined

An invoice reaches
the wrong company

The business process determines the cost of each error

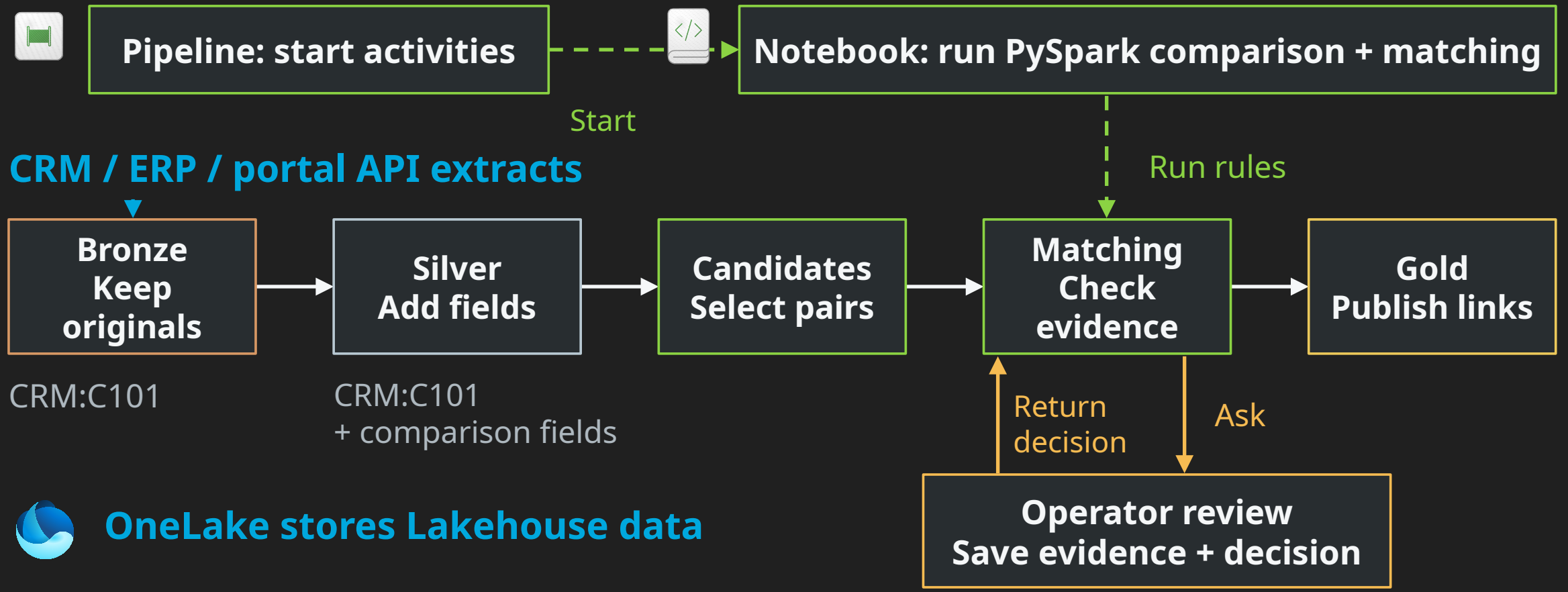
Building the reconciliation flow in Fabric

One record, from source values to a supported result.



Keep the source reference through each step

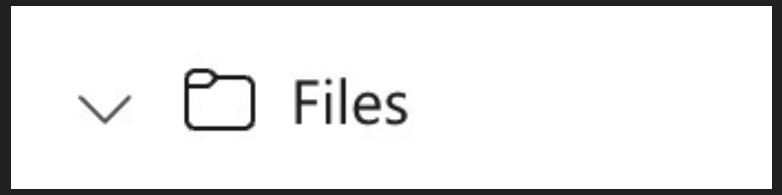
How the records move through Fabric



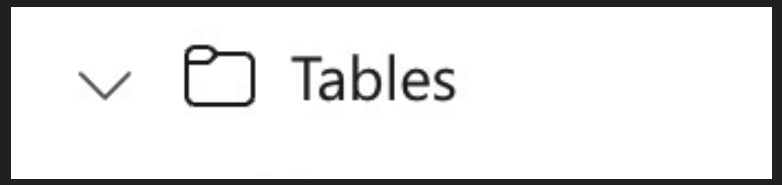
Solid arrows: data and decisions. Dashed arrows: activity coordination.

Bronze keeps the original source records

1. Keep extract files



2. Keep source snapshots



Alfa's Bronze rows

source_key	batch	company_name
CRM:C101	B001	Alfa Ltd
ERP:E501	B001	ALFA LTD

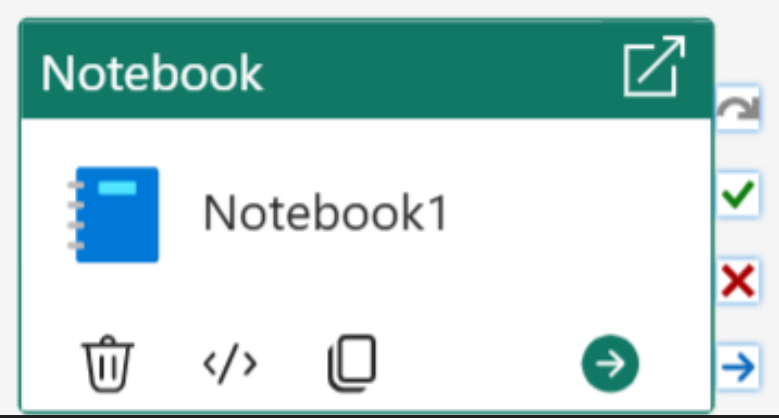
Source + batch + received values

Same batch twice: same snapshot

Microsoft Learn interface close-ups. Originals keep their source keys and batch.

Pipeline starts the selected notebook

1. Add Notebook activity



2. Select the notebook



The notebook to run

Start after the extracts are ready

Pipeline starts it. Notebook runs our rules.

Microsoft Learn interface reference. Notebook and orchestration are configured separately.

Silver adds comparable fields and keeps originals

Source key	Original value	Comparable field
CRM:C101	Alfa Ltd	**alfa ltd**
ERP:E501	ALFA LTD	**alfa ltd**
ERP:E501	Roma St., 12	roma street 12
PORTAL:P901	Roma Street 10	**roma street 10**

Normalisation adds a field. The original stays.

Explicit, limited rules. Source keys and meaningful address differences remain.

Missing and conflicting IDs mean different things

Input	What it tells us	Action
CRM:C103: ID missing	No company-ID evidence	Keep missing
BAD-ID: validation example	Unusable format	Keep original + check
Two usable IDs disagree	Possible identity conflict	Stop automatic link

Two missing IDs do not prove a match

Format validation is separate from verifying the real entity behind an ID.

26 possible pairs become 7 candidates

26 cross-source pairs

7 candidates

Pair	Filter evidence	Candidate?
Alfa CRM:C101 / ERP:E501	Same usable ID: DEMO-001	**KEEP**
Delta CRM:C103 / PORTAL:P902	Postcode 98122 + prefix del	**KEEP**
Nova / Alfa CRM:C104 / ERP:E501	No equal ID or shared block	**EXCLUDE**

Blocking chooses pairs to compare

Local keys and company IDs have different jobs

Identifier	Job	Scope
CRM:C101	Find the original row	CRM namespace
ERP:E501	Find the original row	ERP namespace
Company-ID field	Support company identity	Check source reliability

Assumption: these company IDs are reliable

A valid format does not prove who owns an identifier. Preserve the source keys.

Which field supports the Alfa link?

Field	CRM:C101	ERP:E501
Company name	Alfa Ltd	ALFA LTD
Company ID	DEMO-001	DEMO-001
Address	Roma Street 12	Roma St., 12

Which field supports connecting these records?

Decide the company link before choosing the published field values.

Similar names give us a pair to investigate

Source key	Original name	Company ID
ERP:E503	**Delta Services**	DEMO-003
PORTAL:P902	**Delta Service**	**Missing**

Services / Service: similar text, identity still unresolved

Edit distance measures character changes. Identity still needs evidence.

Aurora: what would you check next?

Field	CRM:C102	ERP:E502
Name	Aurora Consulting	Aurora Consulting
Address	Verdi Street 8	Verdi Street 8

What would you check before linking them?

Names and addresses agree. Is that enough?

Aurora scores 100, but the IDs disagree

CRM:C102

DEMO-002

ERP:E502

DEMO-009

100 points

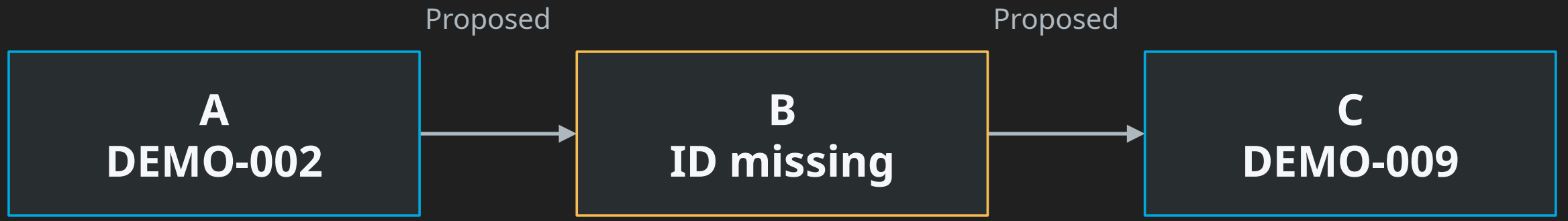
Name 60 + address 25 + postcode 15

KEEP SEPARATE

Reliable ID conflict overrides the score. Points measure field agreement.

Can this record safely join the group?

A proposed group must satisfy every identity check



STOP: DEMO-002 and DEMO-009 conflict

Check every member and previously rejected pairs

A missing ID in the bridge record does not remove the contradiction.

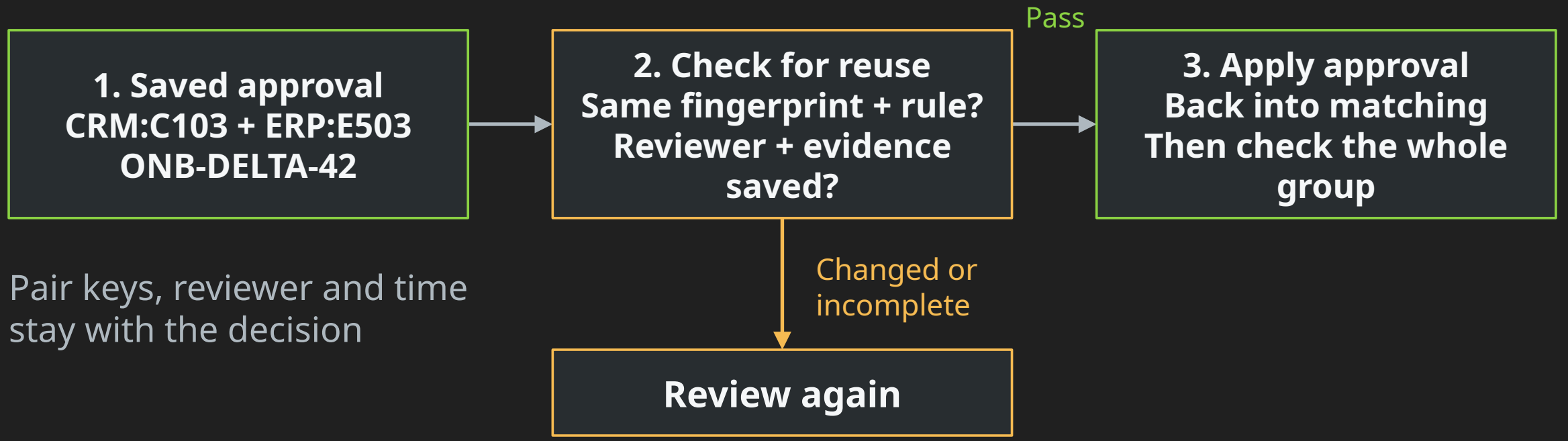
Delta: what evidence would justify this link?

Field	CRM:C103	ERP:E503
Name	Delta Services	Delta Services
Address	Port Street 5	Port Street 5
Company ID	**Missing**	DEMO-003

What extra evidence would justify this link?

Review needs evidence for these exact source keys.

When can we reuse a review decision?



Pair keys, reviewer and time stay with the decision

Fingerprint: summary of comparison fields + rule version

Withdrawn evidence also needs a production flag, even when row fields stay unchanged.

WHEN ENTERPRISE DATA DISAGREES

From decisions to trusted results

A published value must keep the reason for its choice.



Updates can change values or require another decision

Which source would you use for billing?

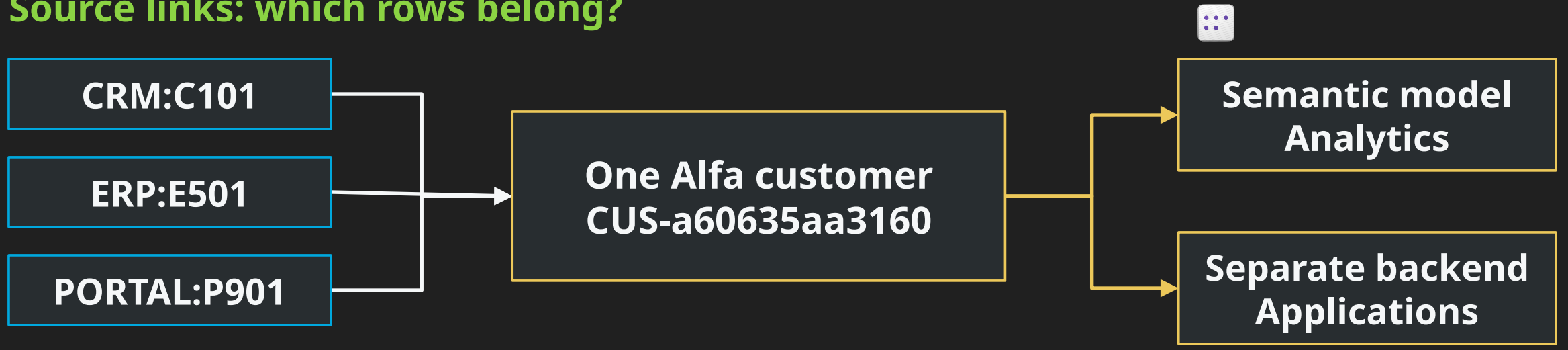
Source key	Company name	Address
CRM:C101	Alfa Ltd	Roma Street 12
ERP:E501	ALFA LTD	Roma St., 12
PORTAL:P901	Alfa	Roma Street 10

Which source would you use for a billing address?

One company can have several address roles. Choose by field and business purpose.

Trace the company — then trace its values

Source links: which rows belong?



Field provenance: why this value?

Published field	Value	Chosen source
Company name	Alfa Ltd	CRM:C101
Billing address	Roma St., 12	ERP:E501

Gold keeps both source membership and the origin of each selected field.

New address, same supported company?

Same source record: ERP:E501



Stable Alfa customer: CUS-a60635aa3160

ID conflict or changed review input: check again

B002: 10 stored versions, 9 current rows. Keep B001; publish the new billing value.

What did we get — and what have we measured?

Company	Company ID	Source links
Alfa Ltd	DEMO-001	3
Aurora Consulting	DEMO-002	1
Aurora Consulting	DEMO-009	1
Delta Services	DEMO-003	2
Nova Travel	Missing	1

5 companies

8 source links

1 held row

PORTAL:P902

Counts check these nine records. Accuracy needs independent labels.

False link: different entities joined

Missed link: one entity stays split

Checked local reference outputs. Include excluded true pairs when measuring candidate coverage.

What changes in production?

Keep the protections. Change how we maintain them.

Current notebook	Production requirement
Group 9 rows on the driver	Distributed grouping + whole-group conflicts
Recompute the current snapshot	Track affected blocks and groups; handle version changes
ID derived from an identity anchor	Persistent identity registry: corrections, merges and splits

A later load does not always mean a newer source value

These production requirements are not features supplied by the current notebook.

Three questions — one practical first step

Are we linking the same kind of object?

Messina: street and address identities differ

What supports or contradicts the link?

Aurora: conflict. Delta: scoped review evidence.

Can we trace the published value?

Alfa: source links + field provenance

Session pack

Notebook + data
Review + expected results
README + sources

Tomorrow: label a small sample and keep the reasons

Local reference checked. The first notebook execution in Fabric still needs validation.

Questions & answers

Which match would your system refuse to make?

Your sources, your evidence, your error costs

8 minutes

Let's connect

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LinkedIn

Thank you. Enjoy the rest of Data Saturday Sofia.